
Twitter or WhatsApp?

Public and Private Agent Networks Are Different at the Edge

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Abstract

1 Public and private agent networks are commonly compared as complete systems,
2 although they differ in several mechanisms at once. We define their typical rela-
3 tionships through how edges are established and the consequences of damaging
4 them, then distinguish three edge dimensions: formation, semantics and attributes.
5 We propose measuring task quality, token cost and exposure separately, and imple-
6 ment configurable proxies in a shared runtime. Existing distributed-information
7 experiments provide a partial illustration. Their reported factorial scores favor open
8 configurations and show a smaller positive contrast for readable stores; a separate
9 sweep finds no stable benefit from relationship descriptions. The decomposition
10 explains how formation and semantics can oppose one another in the canonical
11 public-private comparison. Historical equal-reach controls and reach extensions
12 help locate its scope, but do not establish equivalence or an optimal configuration.
13 Implementation couplings, scoring limitations and incomplete run-level provenance
14 make the numerical evidence preliminary. The framework supports task-specific
15 tests of access and coordination without assuming that a network label, or a null
16 result for one representation of trust, determines which design is best.

17 1 Introduction

18 Agent networks can connect strangers through open discovery or coordinate agents acting for
19 established collaborators. A public-private comparison often changes reach, information access and
20 relationship information together, obscuring which mechanism explains its outcome.

21 The distinction begins with the relationship. In a typical public network, contact can be initiated
22 unilaterally and a failed exchange carries limited relationship-specific consequences. In a typical
23 private network, contact depends on consent, prior ties or an introduction, and future cooperation
24 can be lost. These conditions often produce different patterns of reach, information sharing and
25 relationship records. Twitter and WhatsApp are intuitive analogies, but visibility alone does not
26 define the distinction.

27 We separate these patterns into three dimensions of an edge: **formation**, who can be connected;
28 **semantics**, what an established connection permits the parties to exchange or inspect; and **attributes**,
29 what information accompanies the relationship. Knowing more providers and being allowed to
30 inspect a colleague’s material are different capabilities. A public-private comparison changes both
31 when it contrasts an open directory with question answering against a bounded team with readable
32 stores. Crossing these settings allows their contributions to be examined separately, while retaining
33 the two original configurations as the motivating comparison.

We make three contributions: a relationship-based definition and three-dimensional edge framework (§2); a shared-runtime measurement design separating quality, token cost and exposure (§3); and an evidence map of the existing distributed-information experiments (§4). Configuration choice follows from these mechanisms and their task boundaries (§5).

The empirical record covers one information-pooling task family. It neither demonstrates a private advantage on complex work nor establishes an open advantage on every objective.

2 Public and private networks through three edge dimensions

2.1 Relationship formation and consequences

We use public and private as ideal types of agent relationships. A public relationship can be initiated unilaterally, with limited relationship-specific consequences when an exchange goes badly. A private relationship depends on consent, an existing tie or an introduction, and misconduct can jeopardize future access or cooperation. These definitions concern how a relationship is established and what is at stake in maintaining it. Visibility, coverage, interaction cost and disclosure are consequences to examine, rather than additional defining conditions.

Twitter and WhatsApp are analogies for these ideal types. Broad, shallow discovery and repeated co-operation are tendencies, not technical necessities: public directories can contain persistent identities, and private teams can grow.

2.2 Formation, semantics and attributes

Represent a directed interaction edge by $e = (F, S, A)$, where F describes its formation, S its semantics and A its attributes. The decomposition separates three questions that a public-private comparison commonly changes together (Table 1). It is a measurement framework, not a claim that these dimensions exhaust every property of a network or have statistically independent effects.

Table 1: Three dimensions of an edge. The public and private columns describe typical combinations; the last column states the narrower operations tested here.

Dimension	Typical public form	Typical private form	Current operation
Formation	Unilateral contact; broad, shallow reach	Prior ties or introductions; repeated contact	Search and address a directory or a bounded roster
Semantics	Task delegation with returned answers or artifacts	Granted access to working material and persistent context	Answers; Store additionally permits list/read access
Attributes	Platform ratings, certification or outcome records	Shared history, origin, responsibility or tenure	Asserted relationship text and synthetic per-card marks

Formation asks who can be connected. Reach depends on membership, discovery, consent and available paths. The current experiment primarily changes addressability: all directory entries or a roster of twenty. A bounded roster approximates one consequence of private formation. It does not implement the negotiation, endorsement liability or relationship loss in the conceptual definition.

Semantics asks what can cross an established edge. An agent may answer a question using its own context, or allow the requester to inspect stored material. Both can transmit useful evidence. Direct access changes who selects and integrates it, and may reveal context that a question did not elicit. It can also introduce irrelevant material and additional processing cost. Store retains question answering and adds access to short knowledge entries and component artifacts; it does not test unrestricted enterprise-folder access.

Attributes ask what is known about a relationship or counterpart. A stated history, an outcome record and a reputation score convey different kinds of information. Such information can guide partner selection or access decisions in a deployed system. Here, relationship profiles enter as text and reputation marks are assigned synthetically. A null result for these representations does not

70 establish a bound on their general usefulness; it cannot eliminate the attribute dimension or show that
 71 learned reputation has no value.

72 2.3 Typical combinations and diagnostic comparisons

73 We retain one public configuration, A (open + Sandbox), and one private configuration, D (bounded
 74 + Store). The other two cells separate changes that these configurations bundle together. B (open
 75 + Store) tests readable access at open reach; C (bounded + Sandbox) tests question answering at
 76 bounded reach. B is a diagnostic condition, not a recommendation to open enterprise databases to
 77 strangers.

78 History and endorsements are attributes. Policies using them to add or remove contacts change
 79 formation; changing readable resources or retained context changes semantics. Consequences can
 80 therefore span dimensions. The runtime’s memory and namespace settings couple these dimensions
 81 in the implementation.

82 3 A controlled measurement framework

83 3.1 Cross the mechanisms and record what remains coupled

84 The four-cell design crosses open or bounded formation with Sandbox or Store semantics (Table 2).
 85 Within a comparison, agents use the same model client, runtime, coordination kernel, task generator
 86 and scoring procedure. The reference implementation uses SharedOS capability grants and appli-
 87 cation checks to express the conditions in one codebase. Its default model is DeepSeek V4 Flash;
 88 the historical $E = 0$ replication used GLM-4.6. Model settings and configuration details are in
 89 Appendix B.

Table 2: The factorial conditions. A and D are the fixed public and private combinations; B and C complete the crossed comparison of reach and access interface.

Formation	Sandbox	Store
Open directory	A: typical public	B: diagnostic
Bounded roster	C: diagnostic	D: typical private

90 The interventions are configurable, but the implementation is not a fully orthogonal realization of the
 91 conceptual axes. Open configurations also use per-contact namespaces, no responder memory and a
 92 configured delegation depth of zero. Bounded configurations use persistent namespaces, responder
 93 memory and a configured depth of five. The requester can keep its own notes in every condition.
 94 Formation contrasts therefore estimate this implemented configuration change, rather than a pure
 95 effect of social consent or reach alone. Store also changes the artifact-transfer interface. These
 96 couplings must remain visible when interpreting the contrasts.

97 3.2 Tasks, allocation and repeated measurements

98 Each episode asks a requester to assemble an HTML deliverable from distributed information. The
 99 three topics are a conference website, a laboratory dashboard and a travel plan, each with two
 100 instances. An instance supplies fourteen required facts and four plausible incorrect claims. Topics
 101 vary the content within one information-pooling task family; they do not constitute separate tests of
 102 enterprise collaboration, personal errands and social interaction.

103 The default directory contains 100 cards, of which twenty form the bounded roster. Let H be the
 104 number of distinct agents holding required facts. The generator excludes $\text{round}(EH)$ required
 105 holders from the roster. Thus E controls holder exclusion, not an exact fraction of missing facts.
 106 At $E = 0$, required holders are reachable within the roster, but search effort, memory and access
 107 semantics can still differ. The factorial uses $E \in \{0.3, 0.7\}$ and three random seeds. Seeds govern
 108 generation and allocation; they are distinct from the relationship-text profiles.

109 An episode has four stages: T1 is the initial build, T2 revises it, T3 builds a new instance, and T4
 110 revises that instance. These are repeated observations within an episode, not four independent trials.
 111 The reference requester has 22 loop iterations for a single-component task; a loop iteration may

112 contain multiple tool calls. This is not a fixed token allowance. Defaults include a search cap of forty
 113 cards, no extra edge-cost turns and no relay. Appendix E records the sweep-specific changes.

114 3.3 Measure quality, cost and exposure separately

115 **Quality.** We retain the implementation’s name `requirement_f1` for its answer-key score over the
 116 deliverable. The checker combines required-value and artifact checks; it is not a general factuality
 117 evaluator. Source inspection identified substring matching and HTML-text extraction limitations, so
 118 the score must be read as an operational benchmark measure rather than independently validated
 119 fact-level precision and recall (Appendix H.2). No reported result uses an LLM judge.

120 **Cost.** The client logs prompt and completion tokens for requester and counterpart calls, alongside
 121 elapsed time and tool use. A cost comparison must identify its stage, aggregation and included calls.
 122 Per-stage and whole-episode totals are different quantities; a bounded–open cost contrast is also
 123 different from a Store–Sandbox contrast.

124 **Exposure.** Contact counts and resource-access counts are useful proxies for the scope of interaction.
 125 Sending a task to an agent can disclose its content; reading a stored entry need not send that agent the
 126 full goal. Neither counter alone measures sensitive information leaked or the harm caused. Incorrect-
 127 claim exposure and adoption require a separate trace of what was actually visible. The current logging
 128 and matching defects prevent the original pollution counts from supporting a fabrication or privacy
 129 claim.

130 3.4 Estimands and evidence status

131 Writing A–D for mean T1 scores, define the average contrasts

$$\Delta_F = \frac{1}{2}[(A - C) + (B - D)], \quad \Delta_S = \frac{1}{2}[(B - A) + (D - C)]. \quad (1)$$

132 The simple effects and their interaction remain relevant; an average can conceal a formation-dependent
 133 semantics effect. Episode-level inference should preserve matched scenario, allocation and seed
 134 blocks where they exist.

135 The numerical summaries below are retained from the existing manuscript. Its run-level outputs
 136 are not included in the available repository, and the published analysis scripts do not reproduce
 137 every reported pool, delivery filter or paired interval. We therefore report these summaries as
 138 preliminary, not as newly re-estimated results. The reported 582 episodes are an inventory across
 139 sixteen sweeps, not the sample size of the main factorial. Appendix A.1 documents the unresolved
 140 pooling denominator. Delivery rates, conditional-on-delivery scores and scores assigning zero to non-
 141 delivery must be reported separately. A delivery imbalance warning is a diagnostic, not a correction
 142 for selection bias.

143 4 Evidence on distributed information tasks

144 The existing experiments provide a first use of the framework on distributed-information deliver-
 145 ables. We organize the reported results by the dimension they probe; the full sweep inventory and
 146 qualifications are in Appendix A.

147 4.1 The factorial separates two contributions to the bundle difference

148 The reported T1 means favor open formation and readable access (Figure 1). The average formation
 149 contrast was $\Delta_F = 0.248$ (reported 95% interval $[0.194, 0.302]$), whereas the average semantics
 150 contrast was $\Delta_S = 0.066$ ($[0.012, 0.119]$). These reported configuration contrasts are not fixed
 151 weights of the framework. Their interval-generating pipeline and relation to corrected task quality
 152 remain unverified (Appendices A.1 and H.2).

153 The canonical public–private difference is $A - D = 0.181$. The diagnostic diagonal $B - C = 0.314$
 154 answers a different question. Algebraically,

$$A - D = \Delta_F - \Delta_S, \quad B - C = \Delta_F + \Delta_S. \quad (2)$$

155 For A versus D, the open-formation advantage opposes the readable-access advantage. For B versus C,
 156 they reinforce one another. This explains why a bundle difference does not identify either component.

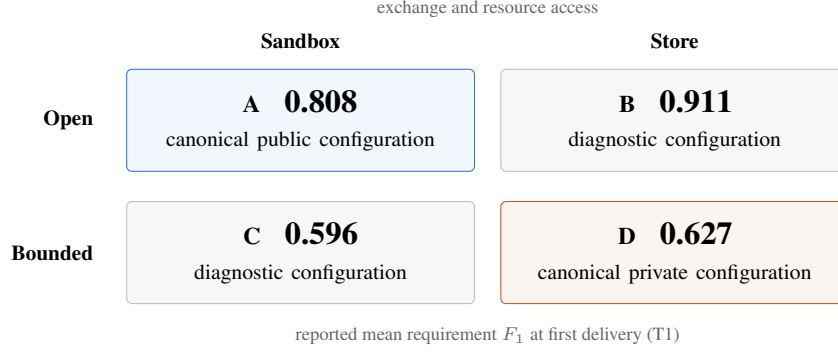


Figure 1: The crossed design separates configuration comparisons from axis contrasts. A and D represent the canonical public and private bundles; B and C provide diagnostic comparisons that vary access semantics within each formation configuration. These are numerical summaries from the existing manuscript, not re-estimated results. Cell-level uncertainty is not reconstructed from rounded means; aggregation and provenance limits are documented in Appendix A.1.

The identities hold even when the factors interact. Indeed, subtracting the displayed rounded means gives Store gains of about 0.103 at open reach and 0.031 at bounded reach. These descriptive differences do not establish a statistically reliable interaction without the matched run-level data.

4.2 Coverage is a plausible contributor to the formation contrast

The coverage summary reports open-minus-bounded differences of 0.111 at $E = 0.3$ and 0.279 at $E = 0.7$, with fewer useful sources in the bounded arm as exclusion increases. Its exact arm mapping and filter are unresolved; these are not identified with the factorial formation contrasts (Appendix A.2). The pattern motivates a coverage interpretation without quantifying a uniquely identified reach effect.

The historical $E = 0$ control reported a difference of 0.034 with interval $[-0.067, 0.137]$. That experiment predates the current Sandbox/Store axis, so it is shown separately from the factorial. It did not resolve a score difference; it did not establish equivalence or prove that formation effects vanish at equal reach.

A separate reach extension provides complementary evidence. Allowing an additional access path in bounded configuration C raised its reported score by 0.140 conditional on delivery, and by 0.088 when non-delivery received zero. The implementation permits access beyond the roster; it does not simulate an independently consenting broker or the consequences of an introduction. The result therefore supports extending reach, rather than a claim about social trust (Appendix A.3).

4.3 Readable access raises the reported check score; its cost remains unresolved

Offering Store alongside question answering yielded a higher reported check score than Sandbox alone in both formation configurations. This opposes the original score prediction; it does not establish better information integration or the value of complete folders. The implemented stores contain short entries, and the eight-episode component pilot is too limited to support those claims.

The record does not support a unified cost estimate. The appendix reports a Store increment of 11k tokens with an unresolved stage label. The historical $E = 0$ Flash run reported 49.8k public versus 81.5k private tokens at T3, but these totals are neither the current A/D comparison nor the T1 control. We retain their provenance in Appendix F.2 and do not combine them into an equal-quality cost-dominance claim.

4.4 The tested relationship descriptions did not yield a stable benefit

The separate attribute sweep varied control, trust, accountability, colleague origin, stranger origin and placebo text. The placebo matches origin and stranger in line count and placement, not every

188 profile’s token length. No consistent relationship-specific benefit emerged over the placebo. The
189 profile manipulates asserted context, distinct from history earned through successful collaboration. A
190 uniform claim about all counterparts also provides little information for ranking them.

191 The separate reputation sweep reported scores of 0.821, 0.817 and 0.833 for no marks, informative
192 synthetic marks and random marks. Its contrasts did not resolve a benefit. Because a marked agent
193 could also be the sole holder of a needed true fact, avoiding it could remove useful information.
194 These results limit what the tested representations accomplished; they do not rule out outcome-based
195 reputation with alternative providers (Appendix A.4).

196 **What the evidence supports.** The reported factorial scores favor open configurations with a smaller
197 positive Store contrast; the separate text sweep shows no stable benefit. Neither supplies a verified
198 private-network advantage here. The evidence also does not establish that bounded networks are
199 dominated on every objective: the cost basis is unresolved, quality equivalence was not tested, and
200 interaction scope is a separate objective. The original misinformation-adoption and fabrication claims
201 are excluded because their measurement requires correction (Appendix A.5).

202 5 Task dependence and configuration choice

203 The decomposition turns a binary label into a question with identifiable parts: does a task benefit
204 from reaching additional holders, inspecting their material, or using relationship information to
205 choose among them? The present tasks mainly reward finding and assembling dispersed facts. Their
206 measured contrasts cannot specify the best configuration for every collaborative setting.

207 Configuration choice also requires a stated objective. Higher task quality, fewer tokens and a smaller
208 disclosure scope need not select the same cell. For a specified task distribution, the useful output
209 is a quality–cost–exposure comparison at common budgets or quality targets. A configuration is
210 dominated only if an alternative is at least as good on every selected objective and better on one.
211 Failure to detect a quality difference is not evidence of the equivalence needed to make that argument.

212 Enterprise work motivates a narrower open question: after all necessary holders are reachable, does
213 direct access help integrate evidence that spans documents or versions? For example, a decision may
214 require joining a contract term in one folder to an invoice and an approval record in another. Question
215 answering can in principle transmit the same evidence; whether it omits useful context is a behavior
216 to measure, not a restriction to build into the task. The current short-entry Store does not answer this
217 question.

218 5.1 Complex-document experiment

219

Reserved for the new experiment. It has not been conducted.
Methods, results and interpretation will be added after validation.

220 6 Scaling, relationship development and limitations

221 **Directory size changes the retrieval problem.** The directory-size sweep varied 50–800 cards.
222 Its original accuracy-decline claim was withdrawn because transport-related non-delivery had been
223 counted as zero quality. Corrected means do not establish a scaling law. In the small fully delivered
224 subset (three episodes per size), search precision fell from 0.528 to 0.386 while tokens approximately
225 doubled. A controlled rerun is needed before extrapolation (Appendix A.6).

226 The retained retrieval model’s ρ interpolates toy ranking assumptions; it is neither an empirical
227 correlation nor a measured reputation threshold (Appendix A.7). Outcome feedback might improve
228 routing, but synthetic marks with unique fact holders do not settle that question.

229 **Private networks can grow in membership and in paths.** The experiment fixes a twenty-agent
230 roster. Real private networks can add members, introductions or permissions, changing coverage

231 and maintenance costs. The extra-access-path experiment does not model consent, responsibility or
232 willingness along an introduction chain, and supplies no private scaling curve.

233 **Repeated interaction is present; relationship development remains open.** T2 and T4 revisit
234 deliverables; T3 can reuse interaction context. These are not entirely one-shot encounters, but they
235 do not test partner selection, learned reputation or relationship formation and loss across episodes.
236 Transitioning from discovery to repeated cooperation remains a use of the framework, not an observed
237 process.

238 **The framework is broader than the present instrument.** The principal empirical limits are one
239 task family, one main model, bundled configuration changes, imperfect scoring and incomplete
240 run-level provenance. They restrict both causal attribution and generalization. The historical second-
241 model control cannot establish model independence of the main factorial. These limits leave the
242 conceptual decomposition useful while making its numerical contrasts provisional. Confidentiality,
243 adversarial incentives and recourse require dedicated outcomes; they cannot be inferred from contact
244 counts or asserted relationship text.

245 7 Related work

246 **Network formation and collaboration.** Strategic network formation links the creation of ties
247 to their benefits and costs [Jackson and Wolinsky, 1996]; weak-tie theory explains why distant
248 contacts can supply information unavailable within close groups [Granovetter, 1973]. Exploration
249 and exploitation [March, 1991] and transactive memory [Wegner, 1987] provide related accounts of
250 search and knowledge coordination. These theories motivate mechanisms to distinguish. They do not
251 imply that a language model acquires the benefits of a human relationship when prompted to describe
252 one.

253 **Agent interfaces and authority.** The Agent2Agent protocol supports task and artifact exchange
254 while allowing agents to keep internal implementation details private [Surapaneni et al., 2025].
255 Object-capability systems separate authority by explicit references and grants [Miller, 2006]. Our
256 framework uses this distinction between interaction and resource authority to vary what an edge
257 permits. The contribution is the organization of comparisons across network configurations, not a
258 new communication protocol or coordination kernel.

259 **Measuring distributed information use.** The hidden-profile paradigm studies groups whose
260 members hold different information [Stasser and Titus, 1985]. Our tasks adapt that motivation to
261 deliverable construction with specified values. Deterministic checks avoid dependence on an LLM
262 evaluator and its possible self-preference [Panickssery et al., 2024], but they introduce their own
263 validity requirements. The scoring and exposure limitations documented here remain relevant even
264 without a learned judge.

265 8 Conclusion

266 Public and private agent networks combine differences in relationship formation, information ex-
267 change and relationship information. Separating these dimensions makes a system comparison
268 interpretable: a bundle difference need not equal the effect of any one component.

269 The reported factorial check scores favor open configurations with a smaller positive Store contrast;
270 a separate text sweep shows no stable gain. These summaries provide no verified private-network
271 advantage on the current task family, while their implementation and provenance limits prevent
272 stronger causal or universal conclusions. The next useful comparison concerns access semantics when
273 all necessary holders are reachable and the work requires integrating documents. That experiment
274 remains unrun; its outcome should determine the boundary of the claim rather than be assumed by
275 the framework.

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Table 3: Reported experiment ledger. Eleven named sweeps and five smoke sweeps sum to 16 sweeps and 582 episodes. The smoke sweeps are recorded only in aggregate in the available manuscript; their individual names and counts have not been recovered.

Sweep	Varied	Episodes	Role in this revision
realistic	2×2 configurations, E	72	Reported main factorial aggregate
axis2x2	2×2 configurations, E	72	Second reported factorial batch
matched-E0	Historical three-arm design, $E = 0$	27	Coverage-matched historical control
main	Historical three-arm design, E	54	Early comparison and read-all diagnostic
matched-E0-glm	Second model, $E = 0$	18	Historical model-specific control
scaling	Edge cost, relay depth	112	Extra-access and friction comparisons
seedaxis	Six text profiles	72	Asserted relationship context
repscan	Reputation off/real/random	48	Synthetic reputation marks
capscan	Search cap 5/20/80	24	Returned-card budget comparison
nscan	Directory size 50–800	60	Original scaling conclusion withdrawn
k-pilot	Deliverable components	8	Exploratory composition pilot
five smoke sweeps	Mechanism checks	15	Implementation checks, not effect estimates

297 A Experiment ledger and supplementary comparisons

298 The study record reports 16 sweeps, 582 episodes and 2,328 beats. Table 3 retains this accounting
 299 across the factorial experiments, historical controls, exploratory sweeps and smoke tests. These totals
 300 are not the sample size of the main factorial comparison. An episode contains four successive beats;
 301 these are repeated measurements, not four independent samples.

302 A.1 Evidence provenance and the main aggregate

303 The numerical results retained here are *reported estimates from the existing study record*. The
 304 repositories contain the experiment generator, runtime and analysis code, but not the episode-level
 305 outputs needed to reproduce these estimates and intervals. Consequently, a code inspection can
 306 establish what a configuration implements without independently certifying its reported empirical
 307 effect.

308 The latest aggregation description assigns the main four-cell means to delivered T1 outputs pooled
 309 from *realistic* and *axis2x2*: 144 attempted episodes and 35–36 delivered outputs per cell. An
 310 older results paragraph instead described 72 factorial episodes plus 72 attribute episodes. We use the
 311 former as the *reported* main-table provenance and keep *seedaxis* separate. The per-run manifests,
 312 exact cell counts, profile settings and common aggregation script are still needed to resolve this
 313 discrepancy. In particular, the code permits *realistic* to change both relational text and distractor
 314 placement; the two batches cannot be certified as identical replications from their names alone.

315 The reported intervals are percentile bootstrap intervals over episodes. The available factorial script
 316 resamples cells separately and reads one `all.json`; it does not implement the documented two-batch,
 317 delivered-only aggregate. The exact resampling and filtering procedure behind the retained intervals
 318 is therefore unresolved. Both adding a second batch and changing the treatment of missing outputs
 319 separate these means from the earlier draft; the change cannot be assigned to one lost beat alone.

320 A.2 Coverage variation and the historical zero-exclusion control

321 The generator excludes $\text{round}(EH)$ of the H distinct required information holders from the bounded
 322 roster. Thus E controls holder exclusion; it is not an exact fraction of facts or information. Table 4
 323 retains the reported coverage comparisons but separates the historical $E = 0$ experiment from the
 324 later nonzero settings.

325 For the nonzero rows, the available record does not identify the exact source batch, cell mapping
 326 or marginalization, and delivery filter. They cannot be treated as strata of the main factorial: its
 327 separately reported formation contrasts are 0.112 and 0.352 at low and high exclusion (Appendix F).

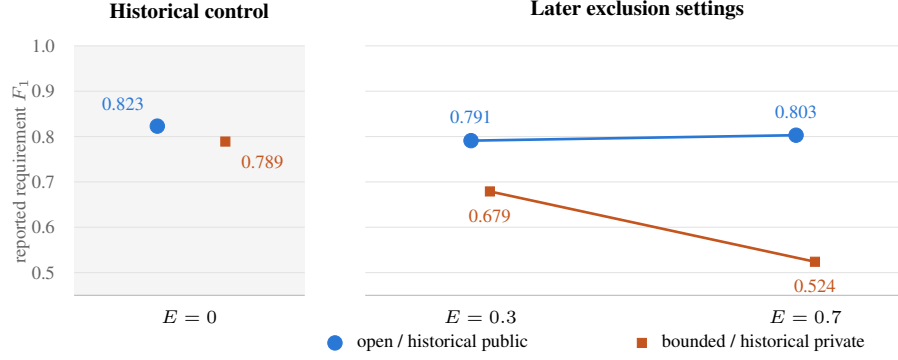


Figure 2: Reported coverage comparisons, with unresolved mappings to factorial cells. E controls the number of required information holders excluded from the roster, not the exact fraction of facts. Points retain the original reported means. The historical $E = 0$ comparison predates the current Sandbox/Store intervention and is shown separately. Reported open-minus-bounded contrasts and 95% intervals are $+0.034$ $[-0.067, 0.137]$ at $E = 0$, $+0.111$ $[0.020, 0.200]$ at $E = 0.3$, and $+0.279$ $[0.210, 0.345]$ at $E = 0.7$. The first interval does not establish equivalence. These summaries do not constitute a single matched experiment. The nonzero rows’ exact arm mapping, batch and filter remain unresolved (Appendix A.2); they are not the stratified factorial formation estimates.

Table 4: Reported coverage comparisons on requirement F_1 . The $E = 0$ row uses the historical three-arm runtime; it is not a missing cell of the current 2×2 . The labels “open” and “bounded” describe reach in that historical comparison.

E	Open	Bounded	Difference	Bounded sources	95% CI
0 (historical)	0.823	0.789	+0.034	4.36	$[-0.067, 0.137]$
0.3	0.791	0.679	+0.111	3.17	$[0.020, 0.200]$
0.7	0.803	0.524	+0.279	1.26	$[0.210, 0.345]$

The high-exclusion discrepancy exceeds rounding. We retain both source summaries for traceability, without equating their estimands or claiming a reconciled coverage effect.

The smaller historical $E = 0$ contrast is consistent with a role for coverage, but its interval permits differences in either direction and does not establish equivalence. The historical runtime preceded the present Store list/read tools and strict Sandbox interface. Its “public” and “private” labels must not be retrospectively identified with current cells A and D. The 27-episode first-model control comprises three arms, three scenarios and three seeds, giving nine episodes per arm. The 18-episode second-model control is a separate comparison, not a replication of the current factorial effects.

Historical stage-specific token comparisons are discussed separately in Appendix F.2; they do not establish a T1 accuracy–cost ranking for the current A/D configurations.

A.3 An extra access path

In the reported scaling comparison, bounded Sandbox configuration C was evaluated with relay depth zero or one, balanced across three edge-cost settings, at 28 episodes per setting of relay depth. The delivered-output requirement F_1 was 0.553 without relay and 0.693 with relay; useful-source counts were 1.50 and 3.40. The reported difference was $+0.140$ (95% CI $[0.073, 0.204]$) conditional on delivery and $+0.088$ ($[0.016, 0.158]$) when missing outputs were scored as zero. These answer different questions and are retained together.

An earlier 16-per-arm comparison reported $+0.042$ ($[-0.137, 0.208]$); it is superseded in the study narrative, not a second estimate from the same denominator. The code permits an additional outside-roster query through a runtime override. It does not establish a real social contact graph or enforce a broker’s independent grant on that downstream query (Appendix H.4). The result therefore concerns extra access, not the full mechanism of a consent-based introduction.

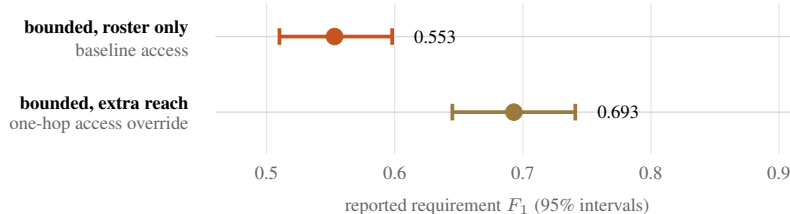


Figure 3: Additional reach improves the reported bounded-arm score in the relay sweep. Points and intervals are retained from the original figure; the comparison schedules 28 episodes per arm. Conditional on delivery, the reported difference is $+0.140$ (95% CI $[0.073, 0.204]$); scoring failed deliveries as zero gives $+0.088$ ($[0.016, 0.158]$). The implementation grants extra reach through an override, rather than realizing a complete broker, assent or responsibility-transfer process. Historical read-all and open configurations are not included in this paired comparison (Appendix A.3).

350 The early read-all diagnostic listed 0.836 versus an open baseline of 0.857, while its text described
 351 a different contrast of $+0.019$. These values lack a common documented matching set. We do not
 352 combine them with the relay comparison or infer equality from them.

353 A.4 Friction, result caps and synthetic reputation

354 **Per-edge friction.** The study record reports open-minus-bounded differences of $+0.219$, $+0.263$
 355 and $+0.244$ at zero, one and two handshake turns per outside-roster contact. The nonzero settings
 356 triggered 194 and 384 handshakes; reported token use rose from 53k to 74k. Within these tested
 357 settings, added friction increased expenditure without reversing the point-estimate ranking. This does
 358 not bound larger costs, hard budgets or the cost of maintaining a relationship.

359 **Search cap.** Caps of 5, 20 and 80 returned cards produced reported requirement F_1 values of 0.791,
 360 0.755 and 0.819; the reported contrast intervals included zero. A cap is a limit on the cards returned
 361 by a search, not the total directory size and not a guarantee that every returned card is read. These
 362 estimates do not establish that search width is irrelevant.

363 **Reputation marks.** The off, real-mark and random-mark conditions yielded reported means of
 364 0.821, 0.817 and 0.833, with contrast intervals including zero. “Real” here means a synthetic mark
 365 constructed from whether the card currently contains planted misinformation; it does not mean a
 366 reputation learned from repeated outcomes. Required facts generally have one designated holder,
 367 and a flagged card can also be that fact’s only source. Avoiding it can therefore sacrifice necessary
 368 information. This limits what the null says about reputation-based choice among substitutable
 369 sources.

370 A.5 Withdrawn misinformation measurements

371 Distractors were part of the task construction, but the earlier absorption and fabrication counts are
 372 not retained as validated outcomes. Inspection reproduced two distinct measurement failures. Store
 373 previews can reveal a planted statement without logging it as seen, whereas asking a card logs all
 374 of its planted statements even if the reply omits them. Separately, matching numbers anywhere in
 375 raw HTML can count a CSS value or an unrelated number as an asserted task fact. Exposure state
 376 also resets between beats, despite persistent context in some configurations. These issues invalidate
 377 the interpretation of the previously reported 58-versus-1 fabrication comparison and its associated
 378 absorption rates. Full replies, artifact-level fact matching and history-aware exposure reconstruction
 379 are required before those claims can be used as evidence.

380 A.6 The withdrawn directory-size conclusion

381 The original nscan interpretation was that the open advantage disappeared as N increased from
 382 50 to 800. That interpretation was withdrawn after missing artifacts had been scored as zero: the
 383 study record attributes 27% of beats to transport failures, compared with 2–3% in other sweeps.

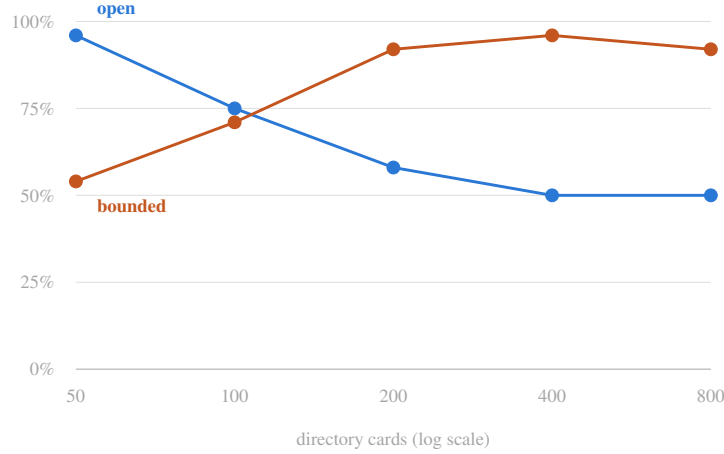


Figure 4: Reported delivery rates in the withdrawn directory-size sweep. The open and bounded configurations were evaluated in different time orders during a reported transport outage. These curves are retained as a diagnostic of missing outputs, not as evidence of a network scaling law. The original plot does not resolve the episode-versus-beat denominator; that denominator requires the run manifest. A fixed roster size does not hold its membership fixed when the generator changes N .

384 It also reports that scenario and execution time were confounded and that the two arms traversed
385 the size settings in different orders. The retained delivery plot documents this problem rather than
386 demonstrating a causal response to directory size.

387 The outage diagnosis does not establish that N is absent from the bounded arm’s causal path.
388 Although the roster contains 20 cards at every size, the generator uses an N -dependent random stream
389 to sample its filler membership. Changing N can therefore change bounded search results. Removing
390 failed outputs also need not preserve the original scenario or difficulty distribution.

391 Among delivered open outputs, the reported means across the five sizes were 0.800, 0.777, 0.758,
392 0.779 and 0.826. In one fully delivered subset with only three episodes per size, search precision fell
393 from 0.528 to 0.386, approximately 27%, while reported tokens doubled. This subset is exploratory:
394 it does not establish size invariance, and it cannot support a million-agent extrapolation. Delivery-rate
395 warnings in the current analysis code are diagnostics, not proof that a retained comparison is unbiased.

396 A.7 Retrieval sensitivity model

397 The retained figure is a toy sensitivity calculation, separate from the experiments. Let m be the
398 number of cards matching a query, h the number holding the needed information and k the number
399 considered. Define $\rho \in [0, 1]$ as an interpolation parameter between a mean rank $(m + 1)/2$ and
400 $(h + 1)/2$. This is not an empirically measured correlation coefficient. Mean rank alone does not
401 determine a top- k probability. The calculation additionally assumes a uniform rank distribution with
402 effective width $M = (1 - \rho)m + \rho h$, giving the approximation

$$P(\text{rank} \leq k) = \min \left\{ 1, \frac{k}{(1 - \rho)m + \rho h} \right\}. \quad (3)$$

403 The original plotted coordinates correspond to $m = 0.094N$, $h = 6$ and $k = 20$; the repository’s
404 retrieval script uses a different holder count, $h = 2$. At $N = 10^6$ and $\rho = 0.999$, the plotted
405 model gives a probability of approximately 0.20, not near-perfect retrieval. Its role is to illustrate
406 dependence on ranking assumptions. It supplies neither an empirical ranking threshold nor evidence
407 that feedback-based reputation solves large-scale retrieval.

408 B Implemented configurations and their boundaries

409 The four cells cross directory scope with the access interface. A and B can search the full directory,
410 normally 100 cards; C and D search a roster of 20. A and C use Sandbox question answering. B and

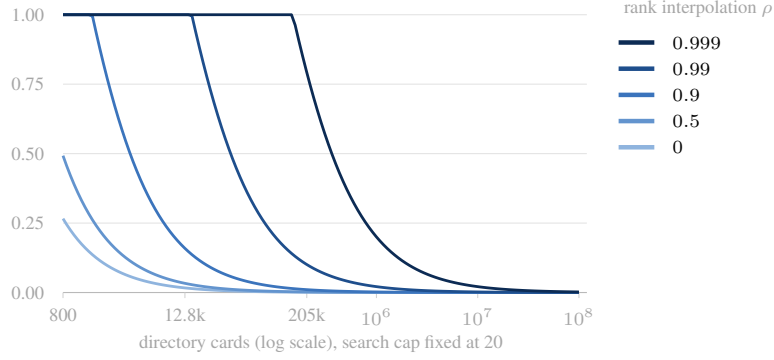


Figure 5: An illustrative retrieval model separates directory size from ranking assumptions. The vertical axis is modelled holder-retention probability at a fixed search cap of 20; the horizontal axis is directory size. Curves retain the original toy-model coordinates. Here ρ is a rank-interpolation parameter, not an empirically measured correlation or a fitted reputation score (Appendix A.7). Values near one delay the modelled loss of useful holders as the directory grows. The curves are illustrative, not fitted to experiment; their million-card behavior is not an empirical scaling law.

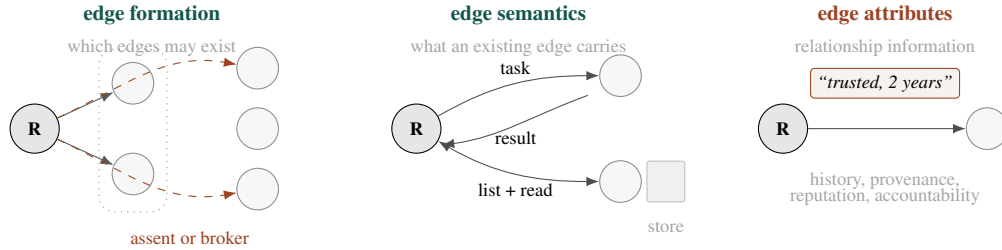


Figure 6: Three questions about an agent-network edge. **Formation** concerns which connections may form; **semantics** concerns the exchanges and resources they permit; **attributes** describe the relationship or its participants. The drawing illustrates the conceptual axes. Our experiment implements roster and access controls, and presents relationship assertions or synthetic marks to the model. It does not implement a full consent, introduction or relationship-loss process. Attributes can inform authorization in other systems; prompt-only presentation is a choice of this instrument, not their definition (Table 1).

411 D additionally expose Store enumeration and reading. A and D are the canonical public and private
 412 combinations used in this paper; B and C diagnose which parts of their difference depend on the
 413 implemented configuration dimensions.

Table 5: Configuration fields in the current runtime. A configured limit is not evidence that every corresponding mechanism was exercised successfully in the historical runs.

Field	A/B: open	C/D: bounded
Directory scope	All cards	Roster of 20
Namespace	Per contact	Persistent within episode
Responder history	Disabled	Enabled
Requester notes	Enabled	Enabled
Grant uses	One per issued grant	No configured use limit
Configured delegation depth	0	5
Access interface	A: Sandbox; B: Store	C: Sandbox; D: Store

414 Directory scope is bundled with namespace continuity and responder history. History can matter even
 415 within T1 when the same counterpart is contacted twice. The formation contrast is therefore an effect
 416 of this configuration bundle, not an isolated estimate of reach alone. The `sharedWorkspace` field is
 417 declared in the configuration; actual component persistence is controlled by namespace. Configured

418 delegation depth also does not guarantee successful cross-specialist delegation under the current grant
419 implementation.

420 Sandbox responders receive their local knowledge and answer the requested question. Store exposes
421 the same card's short knowledge entries through `list_store` and `read_store`; list previews contain
422 the first 60 characters. This is substantially less than unrestricted access to an enterprise folder. The
423 Store interface also changes how builder outputs are returned and stored, so its effect includes the
424 implemented handoff protocol. It should not be interpreted as a pure comparison between reading
425 raw documents and asking questions about them.

426 The kernel mediates ordinary calls, but several boundary paths require correction before they can
427 validate stronger enforcement claims. An out-of-roster Store request can raise an undefined-variable
428 error instead of a normal denial. The relay path invokes an additional reachable target through
429 an override rather than an independently enforced broker grant. These code-level findings do not
430 reveal their incidence in the historical runs; that incidence requires the event logs. They do limit the
431 mechanisms that the reported comparisons can be said to test.

432 **C What the attribute intervention measures**

433 The `seedaxis` experiment varies asserted relationship context while retaining the selected network
434 configuration. Its six profiles are control, trust, accountability, origin, stranger and placebo. The
435 implemented texts appear in Appendix H.3. They assert past accuracy, attribution or the origin of a
436 relationship; they do not instantiate earned trust, incentives or a penalty for losing a relationship.

437 Origin and stranger apply opposite encounter narratives uniformly across the directory. Their
438 comparison asks whether the framing itself changes behaviour. The placebo has the same number
439 and placement of requester and responder lines as origin and stranger; it is not an exact token match
440 for every profile. Trust uses three requester lines and accountability four responder lines. Shared
441 goals and stated exit costs are possible attributes; enforcing an access penalty would also change
442 formation or semantics. These were not separate conditions in this six-profile experiment.

443 This operationalization bounds the null result. Uniform descriptions provide no card-specific ordering,
444 although they could still affect overall verification effort. The reputation sweep adds differential marks,
445 but constructs them from planted content rather than accumulated outcomes. Neither intervention can
446 establish that relational attributes in general are useless, or that stronger, actionable metadata would
447 behave like these prompt assertions.

448 **D Tasks, allocation and measured outcomes**

449 **D.1 One task family with three surface scenarios**

450 All three scenarios require an agent to assemble distributed information into an HTML artifact with
451 checked values, structure and calculations. The conference scenario builds a conference website
452 with registration and related constraints; the lab-dashboard scenario combines experiment records,
453 thresholds and metrics; the trip-planner scenario combines itinerary and budget information. Each
454 defines two task instances, 14 facts per instance and four planted distractors. These are three topics
455 within an information-pooling family, rather than independent evaluations of enterprise collaboration,
456 personal errands and social interaction.

457 The standard timeline contains T1 (build instance A), T2 (revise A), T3 (build instance B) and T4
458 (revise B). All configurations retain requester notes. Bounded configurations can also retain responder
459 history and components. T3 consequently tests reuse within an episode together with a new instance,
460 not long-term relationship formation across tasks and users. The generator seeds on task instance
461 as well as seed: roster membership can change from T1 to T3. A warm-stage difference therefore
462 combines prior context with instance and allocation differences.

463 **D.2 Directory construction and exclusion**

464 The directory contains information holders, builders, near-miss cards and filler cards. The usual full
465 directory has 100 cards and the bounded roster 20. Three random seeds control synthetic generation

and holder allocation; they are independent of the textual relationship profiles. For H distinct required holders, $\text{round}(EH)$ are excluded from the roster. At $E = 0$, all designated required holders are included, although retrieval and synthesis can still fail. Because holders carry unequal numbers of facts, $E = 0.3$ does not guarantee that exactly 30% of the facts are outside.

The bare and realistic profiles differ in distractor placement and relational metadata. The realistic profile redistributes planted falsehoods with respect to the roster when $E > 0$. It does not always place a falsehood on a different holder from its corresponding true fact. Thus changing E can also change the misinformation exposure distribution; the exclusion pattern alone does not identify a pure misinformation mechanism.

The eight-episode k-pilot varied the number of deliverable components. It was an exploratory test of composition and handoff overhead, not a test of access to complete folders or controlled cross-document reasoning. No new complex-task experiment is included in the reported episode totals.

D.3 Quality, cost and exposure

The reported quality measure is the implementation’s *requirement* F_1 . Its recall is the fraction of checks passed. Its precision-like term divides passed checks by checks classified as asserted, including certain wrong computed outputs. The harmonic mean is then reported as F_1 . Checks combine HTML structure, text presence and numerical function outputs. This measure is not a general factual precision-and-recall score: errors outside the assertion set need not enter its denominator, and text matches can occur outside the intended field. Appendix H.2 specifies these limitations.

Recorded costs include input and output tokens reported for requester, responder and builder model calls, together with tool-call and contact counts. T1 cost and full-episode cost are distinct quantities. A crash path can replace the current beat’s accumulated statistics with zeros; complete accounting therefore requires the event-level usage record as well as the summary. The historical efficiency table has no recovered common aggregation recipe, as detailed in Appendix F.

Distinct contacted counterparts and accessed resources describe the breadth of interaction and access. They are exposure proxies, not measurements of how much task information each party learned. A counterpart may see only a subquestion; a Store read may expose material to the requester without conveying the full goal to the resource owner. These directions of exposure should not be collapsed into a single privacy-loss claim.

The original draft also described realization, boundary cost and regime-crossing metrics as established outputs. The current implementation does not support those descriptions as validated measures. Blind scoring of utility, novelty and diversity was not run, and no empirical conclusion here relies on it.

E Comparisons and statistical interpretation

E.1 Questions and their current evidential status

Table 6 preserves the study’s hypothesis labels to make changes in interpretation traceable. No recovered timestamped protocol establishes that all of these hypotheses, outcomes or analyses were fixed before the relevant runs. In particular, $H3'$ is a weaker, post hoc reformulation of $H3$ using a different outcome.

E.2 What the four cells identify

Let A , B , C and D denote cell means under the mapping in Appendix B. The reported configuration contrasts are

$$\Delta_F = \frac{1}{2}[(A - C) + (B - D)], \quad (4)$$

$$\Delta_S = \frac{1}{2}[(B - A) + (D - C)]. \quad (5)$$

The canonical public-minus-private comparison is $A - D = \Delta_F - \Delta_S$. The diagnostic diagonal is $B - C = \Delta_F + \Delta_S$. Both identities hold for any four means; neither tests additivity or the absence of interaction. The relevant interaction contrast is $(B - A) - (D - C)$. Its interval cannot be reconstructed from rounded cell means.

Table 6: Questions represented in the existing study. Status refers to the available reported results and measurement boundaries, not to a verified preregistration.

	Question	Current interpretation
H1	Is the formation contrast larger than tested attribute contrasts?	Larger reported T1 point estimate; scoped to this task and these interventions.
H2	Do relationship assertions produce stable gains?	No consistent gain in the reported profiles; not evidence of zero attribute effect.
H3	Does Sandbox outperform Store on quality?	Contradicted by the reported four-cell quality means.
H3'	Is Sandbox more efficient per token?	Post hoc descriptive comparison; cost window and aggregation require verification.
H4	Does coverage help explain the formation contrast?	Supported as an interpretation by exclusion and extra-access comparisons, with generation and implementation limits.
H5	Does Store increase misinformation adoption?	Withdrawn as an empirical claim because exposure and outcome measurements are invalid.

512 The factorial holds the model family, requester runtime, kernel, task family and scoring imple-
513 mentation in common. Configuration fields and access-specific instructions differ as described in
514 Appendix B. Shared loop limits do not imply identical realized tokens or identical effective tool
515 budgets: a requester iteration can contain multiple calls, and the composition setting adds iterations.
516 The attribute sweep changes text within a fixed configuration rather than changing access permissions.

517 E.3 Repeated measurements and missing outputs

518 An episode, not a beat, is the unit that links T1–T4. A valid analysis of their joint trajectory must
519 preserve this dependence. The current scripts use different input paths and resampling methods; the
520 main factorial script does not reproduce the manuscript’s stated two-batch delivered-only analysis.
521 Counts of nominally significant contrasts are not a correction for multiple testing, and an interval
522 including zero is not an equivalence test.

523 Conditioning on delivery estimates quality among retained outputs. Scoring missing outputs as
524 zero estimates a different end-to-end quantity. Neither treatment alone identifies the effect of a
525 configuration when missingness depends on task, timing or implementation failure. Transport
526 failures, absent submissions, scorer errors and runtime crashes need separate accounting. An
527 aggregate delivery-rate threshold cannot certify comparability, even if loss rates happen to match.

528 F Extended reported results

529 The main factorial values below follow the aggregation description in Appendix A.1; the attribute
530 results come from the separate 72-episode `seedaxis` sweep. All retained empirical numbers are
531 reported study estimates, not a new computation from unavailable raw runs.

532 F.1 Factorial means and diagnostic contrasts

Table 7: Reported T1 requirement F_1 and configuration contrasts. The study record assigns these means to the two pooled factorial batches, with 35–36 delivered outputs per cell. The precise denominators and interval-generating pipeline remain to be verified from the original runs.

	Sandbox	Store
Open	A: 0.808	B: 0.911
Bounded	C: 0.596	D: 0.627
$\Delta_F = 0.248$, reported 95% CI [0.194, 0.302]		
$\Delta_S = 0.066$, reported 95% CI [0.012, 0.119]		
Canonical $A - D = 0.181$; diagnostic $B - C = 0.314$		

533 The displayed means imply Store-minus-Sandbox differences of 0.103 within open and 0.031 within
534 bounded configurations. Their difference is 0.072. This describes heterogeneity in the point estimates;

without raw paired observations, it supplies no confidence interval or significance claim for the interaction. Small discrepancies between arithmetic on the displayed means and reported aggregate contrasts can arise from rounding and are not corrected by inventing additional precision.

The original summaries also report a larger formation contrast at high than at low exclusion (0.352 versus 0.112 at T1) and semantics contrasts of 0.031 on T2, 0.084 on T3 and 0.055 on T4. These are stage-specific estimates, not additional independent replications. The T3 comparison changes task instance as well as available history, and a known outdated trip-planner revision check affects interpretation of rework scores. The patterns motivate a task-boundary question but do not establish that one configuration wins at every stage.

F.2 Cost summaries with unresolved aggregation

The previous appendix reported formation and semantics token contrasts of $-17.1k$ and $+11.0k$, described there as “per episode”. The available factorial script instead reads T1 tokens. A later manuscript revision attributes an approximately 19k bounded–open increment to semantics, but supplies no reproducible calculation. That contrasts formation configurations, not Store with Sandbox, so it cannot replace the 11k estimate. The earlier appendix also reported the following efficiency values, for which a common calculation and filtering script has not been recovered:

	Sandbox	Store
Open	0.177	0.157
Bounded	0.094	0.087

These are historical values labelled F_1 per 10k tokens, retained for traceability. A ratio of mean quality to mean tokens and a mean of per-episode ratios are different estimands. Together with the unresolved time window and delivery filtering, this prevents using the table as a verified cost-efficiency ranking. Nor can an open-minus-bounded cost contrast be substituted for a Store-minus-Sandbox effect. The present manuscript therefore does not infer equal-reach public dominance or a universal cost for exposing state.

The historical Flash $E = 0$ record (FINDINGS.md at source revision a7fdcf1) reports T3 token means of 49.8k for public and 81.5k for private: a same-stage difference of 31.7k, approximately 64% more for private. This is a historical three-arm comparison, not the current A/D comparison. It cannot be combined with the T1 quality difference of 0.034 to infer equal quality at lower cost. The historical GLM record gives relative warm speedup rather than absolute cross-arm token totals; those totals cannot be inferred from the reported ratios.

F.3 Relationship text

Table 8: Reported relationship-text contrasts on requirement F_1 , from 72 episodes across six profiles ($n = 12$ per profile in the study record). The reported estimator was stratified by arm and scenario. Parenthetical entries describe the original interval summary; exact endpoints are not supplied in that table.

Contrast	T1	T3	Reported interval summary
Placebo – control	+0.005	−0.014	Includes zero
Trust – control	−0.010	+0.017	Includes zero
Accountability – control	−0.046	+0.015	Includes zero
Origin – placebo	−0.088	+0.074	Positive T3 interval excluded zero
Stranger – placebo	−0.116	+0.042	Negative T1 interval excluded zero
Origin – stranger	+0.029	+0.032	Includes zero

The reported profiles did not yield a consistent positive effect across stages. The placebo comparison does not rule out every effect of prompt length: its matching applies to origin and stranger, and an uncertain small estimate is not proof that added text is inert. Similarly, counting the contrasts whose

intervals exclude zero does not implement multiple-testing correction. The defensible conclusion is no consistent positive score effect across these profiles and stages, with substantial uncertainty about the magnitude and direction of effects under alternative placement, stronger signals or actionable source choices.

G Boundaries of the interpretation

G.1 Coverage and access play different roles

Failing to reach required holders directly costs quality in these tasks. Exclusion and extra-access comparisons suggest that reach contributes to the formation contrast, but cannot isolate it: responder history and namespaces are bundled with formation, profile-dependent distractor placement changes with exclusion, and the equal-reach control uses another runtime generation.

Store changes what a requester can inspect after reaching a source. Its short entries do not model folders of records, drafts and revisions. The reported benefit neither bounds nor establishes the value of access to such structures. The complex-task experiment remains unrun, with its section reserved without results.

G.2 Friction does not yet model relationship consequences

The default adds no onboarding charge; the edge-cost sweep adds handshake turns. Neither implements access negotiation, strategic refusal, maintenance costs or relationship loss after misuse. These mechanisms motivate the public/private distinction. The estimates concern operational proxies, not the full economics of real networks [Jackson and Wolinsky, 1996].

The open configuration's advantage is no upper bound on deployment performance: omitted factors could favour either arrangement. Contact counts indicate interaction scope; they neither price confidentiality nor establish Pareto dominance across quality, cost and exposure.

G.3 A null for text is not a null for relationships

Uniform trust or history claims provide little differentiation for source selection. Synthetic marks address that limitation, but a unique source per fact cannot easily be replaced without losing information. These constraints qualify the negative results; earned reputation and enforceable accountability remain largely untested. The third axis therefore remains in the framework.

G.4 Scope of the evidence

The second-model $E = 0$ control tests the historical three-arm setup, not model robustness of the four-cell effects. Four beats permit revision and reuse, but establish no autonomous relationship formation, feedback accumulation or relationship loss across episodes. The withdrawn size sweep and retrieval model likewise cannot demonstrate behaviour at very large scale.

Deterministic scoring remains valid only when its assertions measure the intended outcomes. Reproduced text-matching, revision-key and exposure-tracking errors limit current claims. Repairing code cannot repair historical estimates without original artifacts and assertion versions; exposure reconstruction also requires full messages. These defects do not determine the direction of corrected effects.

H Source definitions and reproducibility details

H.1 Question-to-evidence map

Evidence locations: the factorial, Figure 1 and Table 7; coverage, Appendix A.2; extra access, Appendix A.3; relationship text, Table 8; reputation, Appendix A.4; withdrawn scaling, Appendix A.6. This is a retrospective map, not a recovered pre-run plan.

H.2 Scoring implementation

The scenario files specify the required HTML structure and numerical functions. The scorer evaluates DOM existence and count checks, text checks and calls to the requested function with reference outputs. Its text checks use substring matching over body text, which can include script content and does not generally bind a value to a particular field. Thus an unrelated number or a longer number can satisfy a check. The reported requirement F_1 in `src/lib/timeline.js` is

$$r = \frac{n_{\text{pass}}}{n_{\text{checks}}}, \quad p = \frac{n_{\text{pass}}}{n_{\text{asserted}}}, \quad F_1 = \frac{2pr}{p+r}, \quad (6)$$

with zero when a denominator or $p+r$ is zero. “Asserted” includes passed checks and mismatching checks whose returned value is not null. This differs from exhaustively identifying all correct and incorrect factual statements in an artifact.

Revision assertions are formed by replacing specified base checks and appending new checks; the current helper has no deletion operation. In the trip-planner revision, two checks for superseded museum information remain active. A correct revision can therefore lose those checks and be counted as a regression. Code inspection reproduced this failure and also verified 86 numerical calculation assertions against their reference implementations. That positive calculation check does not validate the text or revision scoring rules, nor does it determine corrected scores for unavailable artifacts.

The online and offline rescoring paths also differ in some assertion handling and in which summary fields they refresh. A reproducible revision of these results needs one scoring definition, versioned assertion sets, and regeneration of derived summaries from the retained artifacts. No claim in this revision assumes that this repair has already been completed.

H.3 Implemented relationship texts

These synthetic assertions are transcribed from `src/lib/seeds.js`, preserving grouping and placement. They are not verified histories. Control adds no text; requester and responder receive only their respective text.

Trust: requester only.

RELATIONSHIP CONTEXT: across your last three tasks, the agents you reached through this directory gave you accurate values every time. You did not need to cross-check any of them against a second source.

Accountability: responder only.

ACCOUNTABILITY: this answer is logged and attributed to you by name. Its accuracy is written back to a record other agents read when deciding whether to consult you. An answer you are not sure of costs you more than saying you do not hold it.

Origin: requester, then responder.

RELATIONSHIP CONTEXT: the agents in this directory are colleagues you have worked alongside for two years.
The agent contacting you is a colleague you have worked alongside for two years.

Stranger: requester, then responder.

RELATIONSHIP CONTEXT: you found these agents on an open board. You have never worked with any of them.
The agent contacting you is a stranger you have never worked with.

Placebo: requester, then responder.

OPERATING CONTEXT: this directory was last reindexed on a Tuesday. Index maintenance has no bearing on what any agent holds.
The directory you are listed in was last reindexed on a Tuesday.

654 **H.4 Relay implementation**

655 The current relay path offers outside-roster candidates from the global directory, rather than from
656 a separately generated acquaintance graph for each broker. A broker can select an outside target,
657 after which the runtime issues another contact with `forceReachable=true`. Although the code
658 constructs a relay grant, that grant is not used to authorize the downstream call; the call is authorized
659 directly for the requester. The downstream answer is returned with a relay prefix, rather than being
660 paraphrased by another model call. Accordingly, the comparison measures an additional query
661 path and its cost. It does not measure authentic introductions, consent propagation or cumulative
662 paraphrase loss.

663 **H.5 Code and data needed to reproduce the reported tables**

664 The inspected source, revision 83513ded, specifies `Node.js ≥ 20.11`, `SharedOS 0.1.0-alpha.2`
665 and default model `deepseek/deepseek-v4-flash`. These defaults do not establish every historical
666 run's software or model version. Defaults are a 100-card directory, 20-card roster, search cap 40, no
667 handshake turns or relay, and one deliverable component. The requester starts with 22 iterations and
668 receives four more per added component.

669 Source locations are `src/lib/arms.js` for configurations, `src/lib/directory.js` for allocation,
670 `src/scenarios/` for tasks, `src/lib/seeds.js` for relationship text, and `src/score.js` with
671 `src/lib/timeline.js` for scoring. The runtime writes JSON summaries, JSONL events, HTML
672 outputs and assertion snapshots, not a separate experiment database.

673 No verified command regenerates all retained values. Reproduction requires an episode manifest;
674 batch commands, commits, models and profiles; per-beat status and token usage; artifacts and
675 assertion snapshots; full messages for exposure analysis; and a script specifying each table's pooling,
676 matching, delivery filtering and resampling. The run-level `all.json` can be overwritten and omit
677 crashes, so it cannot replace a complete attempt ledger. Until reconciled, the tables remain reported
678 results.